

JÖNKÖPING UNIVERSITY
School of Engineering

Designing for Feedback: How Microinteractions Influence User Perception in Web Interfaces

Main Subject area: *Informatics*
Author: *Filippa Furman & Lova Konradsson*
Supervisor: *Eveline Ingesson*
JÖNKÖPING 2026 May

This final thesis has been carried out at the School of Engineering at Jönköping University within Informatics. The authors are responsible for the presented opinions, conclusions and results.

Examiner:	Ulf Seigerroth
Supervisor:	Eveline Ingesson
Scope:	15 hp (first-cycle education)
Date:	2026-05-30

Abstract

This study examines how the presence of microinteractions in a web interface influences users' understanding and satisfaction during short, goal-oriented tasks. While microinteractions are widely used in modern digital interfaces, existing research has primarily focused on conceptual discussions rather than controlled empirical comparisons of their effects on user experience.

To address this gap, a quantitative experimental study was conducted using a between-subjects design. Two interactive prototypes of an e-commerce interface were developed: one including microinteractions and one without. A total of 20 participants were randomly assigned to one of the two conditions and completed three predefined tasks. Data was collected through a structured questionnaire measuring understanding and satisfaction, complemented by open-ended responses.

The results indicate that microinteractions have a significantly positive effect on both users' understanding and satisfaction. Participants interacting with the prototype including microinteractions reported clearer interpretation of system feedback and a more positive overall experience compared to those using the static interface. These findings suggest that immediate and context-specific feedback supports users in forming more accurate mental models of system behavior, which contributes to improved user experience.

The study contributes empirical evidence to an area that has previously been dominated by theoretical perspectives and highlights the importance of micro-level interaction design in shaping user perceptions. From a practical perspective, the findings suggest that incorporating microinteractions can enhance clarity and satisfaction in task-based interactions, particularly in web-based environments such as e-commerce.

Keywords: Microinteractions, User Experience, Interface Feedback, Understanding, User Satisfaction, Experimental Design, Human-Computer Interaction

Table of Contents

Abstract ..... **ii**

1 Introduction **2**

1.1 Problem statement 3
1.2 Purpose and research questions 3
1.3 Scope and delimitations 4
1.4 Disposition 4

2 Theoretical framework **6**

2.1 Microinteractions in interface design 6
2.2 Understanding and Satisfaction 7
2.3 Theoretical positioning of microinteractions 8

3 Method and implementation **9**

3.1 Literature review 10
3.2 Data collection 11
3.3 Data analysis 13
3.4 Validity and reliability 14
3.5 Considerations 14

4 Results **16**

4.1 Understanding 16
4.2 Satisfaction 16
4.3 Qualitative results 17

5 Discussion **19**

5.1 Result discussion 19
5.2 Method discussion 21
5.3 Limitations 21

6 Conclusions and further research **23**

6.1 Conclusions 23
6.2 Further research 23

7 References **25**

8 Appendixes **28**

8.1 Appendix 1 – Instruction letter 28
8.2 Appendix 2 – Questionnaire 30
8.3 Appendix 3 – Static prototype 34
8.4 Appendix 4 – Prototype with microinteractions 37

1 Introduction

Digital interfaces are evaluated not only based on visual aesthetics, but also on how clearly and intuitively users can interact with them. User experience (UX) design therefore plays a central role in ensuring that users understand system feedback and can complete tasks efficiently (Norman, 2013). Within UX design, microinteractions refer to small, task-focused interactions such as liking a product, adding an item to a shopping cart, or receiving confirmation feedback after an action (Saffer, 2013). Although microinteractions are widely described in design literature, existing research predominantly focuses on conceptual discussions and design principles rather than controlled empirical comparisons of their effects on users' understanding and satisfaction during short task-based interactions (Saffer, 2013; Kolte & Rao, 2024). Prior work in human-computer interaction has explored how microinteractions and interface feedback influence user perception and interaction behavior, often through observational or applied studies. However empirical research that isolates these effects in controlled, task-based settings remains limited (McDaniel, 2015; Boyd & Bond, 2021).

Users depend on interface feedback to interpret system responses and to understand whether their actions have been successfully completed. When feedback is unclear or insufficient, users may experience uncertainty or confusion, even when interacting with relatively simple tasks (Norman, 2013). Microinteractions have been proposed as a way to support user understanding by providing immediate and recognizable feedback that communicates system status (Saffer, 2013).

Previous research within human-computer interaction has established the importance of constructs such as perceived ease of use and perceived interactivity for positive user experiences (Davis, 1989; Sundar & Kim, 2005). However, while these constructs have been widely examined in UX research, less attention has been given to how specific interface design elements, such as microinteractions, influence users' understanding and satisfaction in controlled experimental contexts.

As microinteractions are widely implemented in digital products, there is a need to further examine their role in supporting the user experience. By empirically comparing interface variants with and without microinteractions, this study aims to determine whether and to what extent microinteractions influence users' understanding and satisfaction during task completion, building on recent empirical investigations of microinteractions in controlled setting (Shwany et al., 2024; Dolić et al., 2024), thus contributing to explanatory knowledge within the field and providing insights that can inform the design of more effective user interfaces.

1.1 Problem statement

In digital systems, UX is a critical factor in interface evaluation, as users expect not only visually appealing designs but also clear and intuitive interactions (Norman, 2013). However, users often face challenges in understanding system feedback and predicting the outcomes of their actions, which can lead to uncertainty, errors, or reduced satisfaction (Saffer, 2013).

One approach to improving users' interpretation of system feedback is the implementation of microinteractions, which provide small, task-focused interface responses during user actions. These elements are commonly applied in modern web interfaces, yet their effects on users' understanding and satisfaction are not fully understood (Davis 1989; Sundar & Kim, 2005).

Recent empirical studies have begun to examine microinteractions in controlled settings (Shwany et al., 2024; Dolić et al., 2024). These studies indicate that microinteractions can influence user engagement and experience in specific interface contexts. However, existing empirical work remains relatively limited and context-dependent, and few studies have isolated the presence versus absence of microinteractions within short, goal-oriented web tasks to specifically examine their influence on users' understanding and satisfaction in a controlled experimental design.

This indicates a knowledge gap: while microinteractions are widely implemented in practice, there is limited empirical evidence on how they contribute to users' comprehension and satisfaction during task completion. Understanding this relationship is relevant for both designers and researchers, as it can inform better interface design strategies that improve usability, reduce confusion, and enhance user satisfaction across a range of digital applications (Norman, 2013; ISO 9241-210, 2019). Previous research in human-computer interaction also emphasizes that subtle interaction cues and interface feedback can significantly shape users' perception of responsiveness and interaction quality (Pibernik et al., 2019; Ma et al. 2025). In addition, clearer insights into how microinteractions influence users' perceptions may contribute to more efficient task completion and support the formation of accurate mental models of system behavior (Norman, 2013; Hestenes, 2006). Investigating these relationships is therefore important not only for practical interface design but also for strengthening the empirical knowledge base within human-computer interaction research.

1.2 Purpose and research questions

Microinteractions are frequently used in modern web interfaces to provide feedback and guide users through interaction processes. Although microinteractions are assumed to improve user experience, there is a limited empirical understanding of how they influence users' understanding and satisfaction when completing short task-based interactions. The purpose of this study is:

To experimentally examine whether and to what extent the presence of microinteractions in a web interface influences users' understanding and satisfaction during task completion.

Rather than treating UX as a single broad construct, the study focuses on two specific and measurable aspects: understanding and satisfaction.

Based on this focus, the following research questions are defined:

[1] How do microinteractions in a web interface affect users' understanding when completing a task?

[2] How do microinteractions in a web interface affect users' satisfaction when completing a task?

1.3 Scope and delimitations

This study is limited to examining microinteractions within a web-based e-commerce context, focusing specifically on short, goal-oriented interactions. The investigation concentrates on how the presence or absence of microinteractions influence users' immediate perceptions during task completion.

User experience is a broad and often described multidimensional construct including both pragmatic qualities (related to usability, functionality, and task performance) and hedonic qualities (related to users' emotional responses, enjoyment, and experimental value) (Hassenzahl, 2010). However, this study deliberately narrows its analytical focus to two selected measurements: understanding and satisfaction, which allows for a controlled comparison of interface variations.

Furthermore, the study examines users' immediate post-task perceptions rather than long-term usage patterns, repeated interactions, or actual purchasing behavior. The findings should therefore be interpreted as reflecting short-term user perception within a controlled setting.

The study is also limited by its sample size and experimental context, which restricts the statistical generalizability of the results. Rather than aiming to produce universally generalizable conclusions, the study seeks to provide insights into how specific microinteractions influence user perceptions in comparable digital environments.

1.4 Disposition

The remainder of this report is structured as follows. Chapter 2 presents the theoretical framework, introducing key concepts related to microinteractions, feedback in interactive systems, and the constructs of understanding and satisfaction.

Chapter 3 describes the methodological approach, including the experimental design, data collection, and data analysis procedures used in the study.

Chapter 4 presents the results of the study, followed by Chapter 5, which discusses the findings in relation to the theoretical framework and previous research, as well as the methodological strengths and limitations of the study.

Finally, Chapter 6 concludes the report by summarizing the main findings and outlining suggestions for future research.

2 Theoretical framework

As outlined in the problem statement, although microinteractions are widely implemented in modern web interfaces, there is limited empirical research examining how their presence influences users' understanding and satisfaction during short, task-based interactions. The theoretical framework focuses on concepts that help explain how interface feedback and interaction design may shape users' perceptions during task completion.

Rather than treating microinteractions as isolated design details, they are operationalized in this study as specific forms of interface feedback. To understand their potential influence, it is first necessary to consider the broader role of feedback in interactive systems, how feedback relates to users' mental models and understanding, and how understanding connects to satisfaction.

Accordingly, the framework begins by discussing the role of feedback in interactive systems and positioning microinteractions as a specific form of such feedback within interface design. It then presents theoretical perspectives related to understanding and satisfaction, which explain how users interpret system responses and evaluate their interaction experience. These perspectives guide the selection of measurement dimensions and form the theoretical basis for interpreting the study's results.

2.1 Microinteractions in interface design

In interactive systems, feedback mechanisms are central for communicating system status and action outcomes to users (Norman, 2013). Prior research in human-computer interaction highlights that timely interface feedback helps users interpret system behavior and supports smoother interaction processes (McDaniel, 2015; Pibernik et al., 2019). When users perform an action, they rely on clear and immediate feedback to determine whether the action has been successful and to interpret the current state of the system. Without such feedback, uncertainty may arise, even in relatively simple task scenarios.

Microinteractions can be understood as one specific form of feedback within digital interfaces. They are small, task-focused interactions that provide immediate responses to users' actions. Typically, they occur as part of a larger workflow but are designed to handle a single, discrete action, such as liking a post, adding a product to a shopping cart, or receiving confirmation after completing a step (Saffer, 2013).

These interactions provide visual and contextual cues that communicate system state and action outcomes. By making the result of an action visible, microinteractions may reduce uncertainty and support users in interpreting what has occurred in the interface. Recent research also highlights how microinteractions contribute to perceived responsiveness and usability by guiding user behavior through subtle but meaningful feedback (Kolte & Rao, 2024).

According to Saffer (2013), loops and modes are elements of microinteractions that support adaptive or repeated interaction patterns. Since the present study focuses on brief, goal-oriented tasks without repeated user engagement, these elements are not included. The emphasis instead lies on immediate and context-specific feedback that occurs during single task sequences.

Despite their widespread use in modern web interfaces, particularly in e-commerce and social platforms (Saffer, 2013), empirical research examining how microinteractions influence users' understanding and satisfaction in controlled task-based contexts remains limited. While recent studies have begun to explore the role of micro-level interaction design and interface feedback in shaping user experience, findings remain fragmented and context-dependent (Boyd & Bond, 2021; Ma et al, 2025). Although some empirical studies have begun to examine users' perceptions of microinteractions (Ma & Chen, 2022), the overall body of experimental research remains relatively limited (Morisso & Ferreira, 2023).

2.2 Understanding and Satisfaction

Feedback plays a central role in how users interpret interactive systems. According to Norman (2013), effective feedback reduces uncertainty by making systems status visible and understandable. When feedback is clear and immediate, users are better able to interpret what has happened and to anticipate what will happen next.

In the context of user experience design, understanding involves how users interpret system responses and form expectations about subsequent actions. This process can be explained through the distinction between conceptual models and mental models (Hestenes, 2006). Conceptual models represent the system's intended logic as designed by the developer, whereas mental models reflect the user's internal representation of how the system works. Understanding in this context occurs when the users' mental model aligns with the system's conceptual model.

In interactive environments, users continuously develop and adjust their mental models based on the feedback and cues provided during interaction (Interaction Design Foundation, n.d.). Clear and structured feedback therefore supports users in forming more accurate interpretations of systems behavior (Norman, 2013; Sutcliffe & Hart, 2016). When action outcomes and state changes are explicitly communicated, users are more likely to interpret the interface correctly and proceed confidently in task completion.

In this study, understanding is defined as how well participants can interpret system responses and anticipate what will happen next during task completion, based on the alignment between their mental models and the system's conceptual model (Hestenes, 2006; Norman, 2013). This operationalization allows understanding to be measured in relation to the specific interface feedback provided in the experiment. From this

perspective, feedback mechanisms embedded within an interface may influence the extent to which users experience understanding and satisfaction during interaction.

User satisfaction reflects users' overall evaluation of an interactive experience and is recognized as a core dimension of usability alongside effectiveness and efficiency (ISO 9241-210:2019). Satisfaction concerns how positively users perceive the interaction process, not only whether the task was successfully completed.

Empirical research indicates that interfaces that are easier to interpret and navigate tend to generate higher satisfaction ratings (Saputra et al., 2024). When users experience clarity and predictability in their interaction, they are more likely to evaluate the experience positively. Understanding can therefore be viewed as closely related to satisfaction, as clearer interpretation of system behavior may contribute to a more favorable overall assessment of the interaction.

2.3 Synthesis of the Theoretical Framework

From the theoretical perspectives outlined above, microinteractions can be conceptualized as discrete feedback mechanisms that communicate system state and action outcomes during task completion. By providing immediate and context-specific signals, microinteractions may support users' understanding of system behavior. Since understanding is closely related to users' evaluative experience, microinteractions may also influence satisfaction.

However, while prior research has established the importance of feedback and mental model alignment in interactive systems, fewer studies have experimentally isolated microinteractions as specific feedback elements in short, task-based web contexts. This study therefore investigates whether the presence or absence of microinteractions influence users' understanding and satisfaction during goal-oriented interaction.

Based on the reviewed literature, this study assumes that microinteractions function as interface feedback mechanism that make system states visible (Norman, 2013). This feedback support users in forming accurate mental models, thereby improving understanding (Sutcliffe & Hart, 2016). Improved understanding is, in turn, associated with higher user satisfaction (Saptura et al., 2024). These relationships form the analytical basis for interpreting and discussing the study's results.

3 Method and implementation

The purpose of this study is to investigate how the presence or absence of microinteractions in a web interface influences users' understanding and satisfaction during short, goal-oriented tasks. To examine this relationship, a quantitative experimental design was applied. Experimental designs are commonly used in human-computer interaction research to investigate causal relationships by systematically manipulating variables under controlled conditions (Lazar et al., 2017; Pallant, 2020). The study employed a between-subjects experimental design, in which participants were assigned to one of two conditions and interact with only one version of the interface. A between-subjects design is appropriate when comparing different interface variations, as it reduces the risk of learning effects and carryover effects that may occur if participants are exposed to multiple conditions (Lazar et al., 2017; Pallant, 2020).

To fulfill its purpose, the study compared two interface prototypes, one with microinteractions and one without, examining how these variations influence users' perceptions while performing a defined task.

The study is conducted as a controlled experiment using a between-subjects design. Participants were assigned to one of two conditions, either a prototype with microinteractions or a version without. The independent variable was the presence or absence of microinteractions. The dependent variables were the users' understanding and satisfaction, measured through post-task questionnaire responses.

By manipulating only the presence of microinteractions and keeping all other design elements consistent, the study follows principles of controlled experimental design. This approach strengthens internal validity by ensuring that observed differences can be attributed to the independent variable rather than external factors (Lazar et al., 2017; Pallant, 2020). The microinteractions evaluated in this study were small, task-focused interface elements that provided immediate, context-specific feedback to the user. Specifically, these included liking a product (see Figure 1), adding an item to a shopping cart (see Figure 2), and navigating to checkout, where the version with microinteractions included an additional shortcut for progressing to the checkout page.



Prototype without microinteraction *Prototype with microinteraction*

Figure 1. Comparison of the "like" interaction in the two interface conditions.

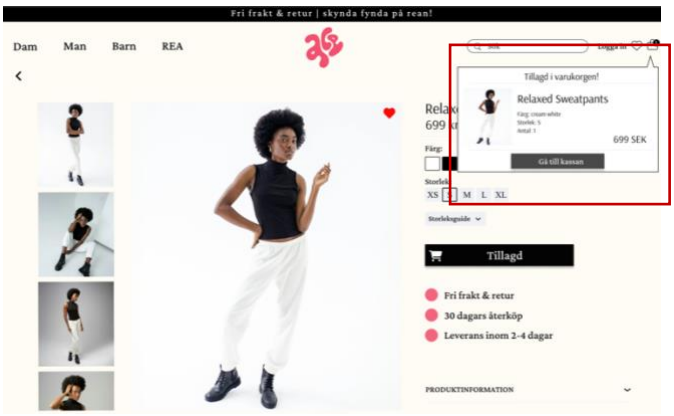


Figure 2. Comparison of the two interface conditions during the shopping cart interaction. The highlighted elements illustrate where microinterctions provide visual feedback.

Each microinteractions were designed to support discrete steps within the task, enabling the study to examine how interface feedback influences users' understanding and satisfaction. All other interface characteristics remained constant to strengthen internal validity. The full prototype designs are presented in Appendix 3 (without microinteractions) and Appendix 4 (with microinteractions).

A quantitative approach was considered appropriate, as the objective was to measure and compare participants' understanding and satisfaction across two experimental conditions (Pallant, 2020). The collected data allowed for statistical comparison between the groups by analyzing mean scores on the understanding and satisfaction scales.

To complement the quantitative data, open-ended questions were included, allowing participants to elaborate on their experiences and provide additional context to their ratings. This meant that both quantitative and qualitative analyses were conducted. However, the primary analytical focus remained on the quantitative comparison between experimental conditions, while the qualitative analysis was used to support and contextualize the quantitative findings.

3.1 Literature review

The literature review was conducted to identify existing research on microinteractions, user feedback in web interfaces, and their influence on understanding and user satisfaction. Google Scholar and ResearchGate were primarily used due to their broad coverage of peer-reviewed research in human-computer interaction and UX design.

Searches were conducted using combinations of keywords including: "microinteraction", "user satisfaction", "perceived understanding", "usability", "human-computer interaction", "task-oriented interaction", "user perception",

"interactive systems", "experimental design in HCI". Example of search strings included: "microinteraction AND user satisfaction", "interface feedback AND usability" and "experimental design in HCI AND user experience".

Searches were conducted using combinations of keywords in Google Scholar and ResearchGate. Around 60 titles and abstracts were screened for relevance to microinteractions, user perception, and experimental interface studies. 25 articles were selected for full-text review, of which 13 peer-reviewed publications were included in the theoretical framework based on relevance and methodological quality.

3.2 Data collection

Data was collected through an experimental procedure involving two interactive prototypes developed in Figma. Figma is an interface design and prototyping tool commonly used in UX development (Figma, n.d.). The data collection procedure can be understood as a form of usability testing, where participants perform predefined tasks using an interface prototype while their perceptions are measured. Usability testing is a widely used method in UX research to evaluate how users interact with systems and to assess their experience during task completion (Rubin & Chisnell, 2008; Lazar et al., 2017).

The prototypes represent a realistic e-commerce web interface designed to simulate a short, goal-oriented task scenario. Both versions were identical in layout, structure, content, and task flow. The only difference was the presence or absence of microinteractions. These microinteractions include visual feedback, task progression indicators, and immediate confirmation of user action, designed to guide users and provide clear and timely feedback. The task scenario consists of three sequential actions: marking a product as a favorite, adding it to the shopping cart, and navigating to the cart page. The selection of three interaction points was intended to create a controlled and focused task scenario, allowing the effects of microinteractions to be isolated while minimizing the influence of external variables. While this limited number of interactions may not capture the full complexity of real-world usage, it enables a clearer examination of the relationship between microinteractions and users' understanding and satisfaction. These actions reflect typical goal-oriented steps in an online shopping process and enable observation of user perceptions during discrete interaction moments. The images used in the prototypes were obtained from Pexels, a free stock image platform, and were used for illustrative purposes only (Pexels, n.d.).

A total of twenty participants were recruited for the study, aligning with recommended sample sizes for usability testing in similar contexts (Macefield, 2009). Participants were recruited through convenience sampling, including students at the university for in-person testing and friends and acquaintances contacted via email for remote participation. They were divided into two groups, with ten participants testing the version that includes microinteractions and ten participants testing the version without microinteractions. Each participant completed the same predefined task within the interface using the same task in both conditions is ensuring comparability between

groups and reduce the influence of external variables. Participants were randomly assigned to the two interface conditions to reduce selection bias and strengthen comparability between groups.

Data collection was conducted in both in-person and remote settings. Seven participants completed the test in person, while thirteen participants participated remotely. Although efforts were made to standardize the procedure across both settings, differences in testing environments may have influenced participants' experiences. Six of the participants were design students from the same academic program as the researchers. These participants were evenly distributed across the two experimental conditions, with half assigned to the prototype with microinteractions and half to the version without microinteractions. This distribution was intended to reduce potential bias related to prior design knowledge across experimental conditions.

After completing the task, participants responded to a structured questionnaire administered through Google Forms. The questionnaire was administered in Swedish, as this is the participants' native language. Using the participants' mother tongue is expected to reduce the risk of misinterpretation and increase the reliability of their responses, particularly when measuring subjective perceptions such as satisfaction. Once all questionnaires were collected, the responses were translated into English and compiled for analysis.

The questionnaire consisted of items measuring both understanding and satisfaction. Satisfaction was assessed using items adapted from the USE (Usefulness, Satisfaction, and Ease of Use) questionnaire after task completion (Lund, 2001). These items were presented as Likert-scale statements, where participants rated their agreement with statements related to their experience of the interface. Responses were measured using a five-point Likert scale ranging from strongly disagree (1) to strongly agree (5). A five-point scale is considered suitable for this study, as it provides sufficient granularity in responses while remaining clear and easy for participants to understand.

Understanding was measured through a small number of evaluation questions focusing on how well participants felt they were able to interpret system feedback and anticipate the next step during the task. These questions were designed to capture participants' subjective comprehension of the interface interaction. While measuring understanding through self-reported data has limitations and may not fully reflect actual comprehension, such measures are commonly used in user experience research to assess users' perceptions. Objective performance-based measures, such as task success or error rates, were considered but not included in this study, as the focus was on the experimental aspects of interaction. Future research could benefit from combining both subjective and objective measures.

The data collection procedure was standardized to ensure that all participants receive the same instructions and complete the task under similar conditions. Specifically, all participants were given identical written and verbally communicated instructions before beginning the task. These instructions include information about the purpose of the

study, task procedure, and participants' rights, such as voluntary participation and anonymity. The full instruction script is provided in Appendix 1.

All participants performed the test on a computer and interacted with only one of the two prototypes. Conditions were kept consistent across participants within the same group. This consistency is intended to strengthen the internal validity of the study and ensure that any observed differences between groups can be attributed to the manipulated variable. Participants were not informed about the specific focus on microinteractions prior to the test, in order to avoid influencing their responses.

In addition to the structured items, a small number of open-ended questions were included to allow participants to describe their experiences in their own words, providing more nuanced insights into the reasons behind their ratings and complementing the quantitative data. Responses to open-ended questions were analyzed thematically to identify recurring patterns that contextualize the quantitative findings.

3.3 Data analysis

The data collected from participants were analyzed quantitatively to investigate how the presence of microinteractions in web interfaces influences users' understanding and satisfaction. Independent samples t-tests were used to compare responses between the two groups: participants interacting with a prototype with microinteractions, and participants using a prototype without microinteractions. The independent samples t-test is commonly used to compare mean differences between two independent groups in experimental studies (Pallant, 2020). This test helped to determine whether any observed differences in user experience are likely due to the interface design rather than random variation.

Thematic analysis was applied to open-ended responses to identify common categories that support interpretation of quantitative results. Thematic analysis is a widely used method for analyze qualitative data by identifying and interpreting patterns withi participants' responses (Braun & Clarke, 2006). The qualitative findings were used to complement and reinforce the quantitative results by providing insight into participants' experiences and helping to explain the underlying reasons for the observed differences in understanding and satisfaction.

A significance level of 0.05 was applied in all quantitative analyses. This indicates that the observed results would be unlikely (less than 5% probability) under the assumption that there is no true effect of the manipulated variable (Maze, n.d.; Pallant, 2020). Choosing this standard ensures that the study minimizes the risk of Type I errors, i.e., falsely concluding that microinteractions have an effect when they do not, while maintaining a rigorous and transparent approach to analysis.

In addition to statistical significance testing, effect sizes were reported to indicate how meaningful the observed differences are in practice, not only statistically. Reporting effect sizes together with p-values allows the results to be interpreted not only in terms

of statistical significance, but also in terms of their practical impact on user experience. This combination provides a clearer understanding of how microinteractions affect users' perceptions during task-based interactions.

The analysis was guided by the constructs outlined in the theoretical framework: understanding and satisfaction. By linking these measurements to interface variations, the study aims to provide evidence-based insights into how microinteractions influence user perceptions during short, goal-oriented tasks. The combination of descriptive and inferential analysis, together with clearly defined significance criteria, supports a transparent and replicable methodological approach.

3.4 Validity and reliability

To ensure that the study produces meaningful and trustworthy results, attention was given to both validity and reliability throughout the research process. Validity refers to whether the methods and instruments measure what they are intended to measure. In this study, previously validated questionnaires addressing satisfaction were used (Lund, 2001), adapted as necessary for the context of the experiment and administered via Google Forms in Swedish. Using established instruments strengthens the validity of the measurements, as these questionnaires have been tested in prior research and shown to reliably capture the intended constructs.

Reliability refers to the consistency of measurements. The survey was structured and standardized so that all participants receive the same instructions and complete the same tasks under similar conditions. This approach ensures that the data collected is consistent across participants and that repeating the study with a similar sample would likely produce comparable results. By maintaining a controlled setup and clear written instructions, the study strengthens the reliability of the collected data and the credibility of the findings.

In addition, the reliability of the qualitative analysis was supported by following a structured thematic analysis procedure in line with Braun and Clarke (2006). The analysis process was conducted systematically and documented to ensure transparency and consistency in how themes were identified and interpreted. This structured approach reduces the risk of subjective bias and increases the trustworthiness of the qualitative findings.

In combination, careful attention to validity and reliability provides a solid foundation for analyzing how microinteractions influence users' perceived experience. Documenting these methodological decisions clearly also ensures transparency, allowing others to understand and potentially replicate the study.

3.5 Considerations

Throughout this study, several considerations guided the research process to ensure it is conducted responsibly and effectively. Scientific considerations were addressed by following established methodological standards in UX and human-computer interaction research (Lazer et al., 2017), including careful design of tasks, controlled comparison

of prototypes, and systematic data collection. These steps ensure that the study produces credible and meaningful results that contribute to the understanding of microinteractions in web interfaces.

Ethical considerations are central to the study. All participants received clear information about the purpose of the study and the procedures involved. Informed consent was obtained from all participants prior to the study, as well as their right to withdraw at any time without consequences. Participants' data was anonymized, stored securely, and used solely for the purposes of this research. Ethical considerations were also guided by established research principles, such as those outlined in the Swedish Research Council's *Good Research Practice 2024*, which provides orientation in ethical requirements and supports reflection on ethics throughout the research process (Swedish Research Council, 2025).

Practical considerations include the feasibility of conducting the study with a limited number of participants and within the timeframe of the thesis. The study focuses on short, task-based interactions in a controlled environment to ensure that data collection is manageable while still providing insights into user experience. Any limitations related to sample size, experimental setup, or prototype design were clearly documented to ensure transparency and allow for appropriate interpretation of the results.

By addressing these considerations systematically, the study aims to maintain scientific rigor, ethical responsibility, and practical feasibility, while producing results that are trustworthy and relevant to both researchers and practitioners in UX design.

4 Results

This chapter presents the results of the study. First, the collected data is presented, followed by the results of the statistical analysis based on the methods described earlier, and finally the results from the thematic analysis. The results are structured according to the study's two variables: understanding and satisfaction.

4.1 Understanding

Participants in the group using the interface without microinteractions (n = 10) reported a mean understanding score of 2.68 (SD = 1.31). Participants in the group using the interface with microinteractions (n = 10) reported a higher mean understanding score of 4.50 (SD = 0.75). An independent samples t-test was conducted to compare understanding between the two groups. The results showed a statistically significant difference, $t(18) = -3.81$, $p = .001$, with a 95% confidence interval of [-2.82, -0.82]. The effect size was large (Cohen's $d = 1.71$).

The result shows a statistically significant difference in understanding between the two interface conditions.

4.2 Satisfaction

Participants in the group using the interface without microinteractions (n = 10) reported a mean satisfaction score of 2.43 (SD = 1.46). Participants in the group using the interface with microinteractions (n = 10) reported a higher mean satisfaction score of 4.20 (SD = 1.16). An independent samples t-test was conducted to compare satisfaction between the two groups. The results showed a statistically significant difference, $t(18) = -3.01$, $p = .008$, with a 95% confidence interval of [-3.01, -0.54]. The effect size was large (Cohen's $d = 1.35$).

The result shows a statistically significant difference in satisfaction between the two interface conditions.

An overview of the results is presented in Table 1.

Table 1

Table 1 presents a summary of the results from independent samples t-tests.

Variable	**Group 1 - Without microinteractions (n = 10) M (SD)**	**Group 2 - With microinteractions (n = 10) M (SD)**	***t***	***df***	***p***	**Cohen's *d***
Understanding	2.68 (1.31)	4.50 (0.75)	-3.81	18	.001	1.71
Satisfaction	2.43 (1.46)	4.20 (1.16)	-3.01	18	.008	1.35

4.3 Qualitative results

To complement the quantitative findings, a thematic analysis was conducted on participants' responses to the open-ended questions. The analysis followed the approach outlined by Braun and Clarke (2006), resulting in the identification of four main themes related to users' experiences of the interface. To provide a clearer comparison between the experimental conditions, the themes are presented in relation to participants' experiences with and without microinteractions.

Theme 1 – Clear and immediate feedback supports understanding (observed in the condition with microinteractions)

Participants frequently described how visual feedback, such as pop-up notifications, hover effects, and changes in icons, made it easier to understand what actions had been performed. For example, several participants highlighted that confirmation messages when adding items to the cart or marking them as favorites contributed to a clear and smooth interaction experience. One participant stated:

"When I liked or added a product to the cart, a small pop-up appeared which made it clear what had happened."

Theme 2 – Lack of feedback creates uncertainty (primarily observed in the condition without microinteractions)

In contrast, participants who did not receive clear feedback reported uncertainty about whether their actions had been successfully completed. Some participants described repeating actions multiple times due to absence of confirmation, indicating that insufficient feedback negatively affected their understanding of the system. One participant that conducted the prototype without microinteraction explained:

"I repeated the same actions several times because I did not understand if the website responded to my actions."

Theme 3 – Familiar design supports navigation and understanding (observed across both conditions)

Several participants noted that the interface resembled other e-commerce websites, which made it easier to understand how to interact with it. Familiar elements such as icons, layout, and navigation structure contributed to a more intuitive user experience. For example, one participant that conducted the prototype without microinteractions stated:

"It looked like a typical web shop, which made it easy to understand how to use it."

Theme 4 – Visibility of system status supports understanding (more prominent in the condition with microinteractions)

Participants also emphasized the importance of being able to visually confirm system status after performing an action. Changes in icons or indicators, such as a highlighted favorite icon or updates in the shopping cart, were described as important for understanding whether an action had been successfully completed. One participant that conducted the prototype without microinteraction explained:

"I would have liked the heart icon to change so that I could clearly see that the product was marked as a favorite."

5 Discussion

The purpose of this study was to examine whether and to what extent microinteractions in a web interface influence users' understanding and satisfaction during task completion. The results showed that participants who interacted with an interface with microinteractions reported significantly higher levels of both understanding and satisfaction compared to those who used an interface without microinteractions.

The discussion interprets the findings based on the theoretical assumption that microinteractions function as feedback mechanisms (Norman, 2013) that influence understanding (Sutcliffe & Hart, 2016), which in turn affects satisfaction (Saptura et al., 2024).

5.1 Result discussion

Regarding the first research question, the findings indicate that microinteractions have a positive effect on users' understanding. Participants in the condition with microinteractions demonstrated significantly higher understanding scores, suggesting that immediate and context-specific feedback helps users interpret system responses and anticipate subsequent actions. This finding aligns with Norman (2013), who emphasizes the importance of clear feedback in enabling users to understand system behavior. These findings are also supported by the qualitative results, particularly Theme 1 and Theme 4, where participants described how visual feedback and visible system status made it easier to understand whether actions had been successfully completed.

In addition, participants in both conditions described how familiar design elements supported their understanding of the interface (Theme 3). However, despite this baseline familiarity, participants interacting with the version containing microinteractions still reported significantly higher understanding scores. This may indicate that microinteractions provide additional support beyond familiarity with common interface conventions.

By providing visual and responsive cues, microinteractions appear to reduce uncertainty and support users in forming more accurate mental models of the interface, as described by Hestenes (2006). This interpretation is further reflected in Theme 2, where participants in the condition without microinteractions described uncertainty regarding whether their actions had been registered by the system, sometimes leading them to repeat actions multiple times. When the system's conceptual model is clearly communicated through feedback, users are better able to align their mental models with the system, leading to improved understanding during task completion.

With regard to the second research question, the results show that microinteractions also have a significant positive effect on user satisfaction. Participants who interacted

with the interface with microinteractions reported higher satisfaction levels than those in the control condition. This suggests that immediate feedback not only supports cognitive processes related to understanding, but also enhances the overall user experience. This finding is consistent with previous research indicating that ease of interpretation and interaction feedback contribute to higher satisfaction (ISO 9241-210:2019; Saputra et al., 2024). Microinteractions may contribute to a sense of responsiveness and control, which are key factors in positive user experiences.

The large effect sizes observed in both analyses further indicate that the differences between the groups are not only statistically significant but also practically meaningful. This strengthens the argument that microinteractions play an important role in shaping user experience, even in short, task-based interactions. The findings therefore support the assumption that small, task-focused design elements can have a substantial impact on how users perceive and interact with digital interfaces.

An additional observation during the study was that participants with a background in design appeared to interpret and navigate the interface differently compared to participants without such experience. Specifically, participants with design experience tended to be more sensitive to the absence of feedback and more critical of missing or unclear microinteractions, while participants without such experience relied more on trial-and-error during task completion. While this difference was not formally tested, it may indicate that prior knowledge of interface design influences how users perceive and evaluate microinteractions. Some variation in participant responses was also observed, but these differences were not systematically analyzed due to the scope of the study and were therefore treated as contextual observations rather than formal findings.

In contrast, a few participants without a background in design reported relatively high levels of satisfaction even in the condition without microinteractions, deviating from the overall pattern observed in the results. This may suggest that these users perceived the interface as functioning as expected, without relying on additional feedback elements. For such users, a simpler interface without microinteractions may still be interpreted as clear and sufficient for task completion.

Together, these observations indicate that user experience is not uniform and may be influenced by individual differences such as prior experience, domain knowledge and expectations of system behavior. This aligns with previous research suggesting that users' mental models and prior exposure to similar systems shape how they interpret interface feedback and evaluate their experience (Norman, 2013; Hestenes, 2006). These findings highlight the importance of considering user background when designing and evaluating interactive systems.

Overall, the results suggest that microinteractions function as an effective mechanism for communicating system status and guiding user behavior. By making interaction

outcomes more explicit and predictable, microinteractions support users in navigating the interface. These findings contribute to the existing body of research by providing empirical support for the role of microinteractions, an area that has previously been dominated by conceptual and design-oriented discussions.

5.2 Method discussion

The methodological approach of this study presents several strengths. The use of a controlled experimental design made it possible to isolate the effect of microinteractions by keeping all other interface elements constant. This strengthens the internal validity of the study, as differences in outcomes can more confidently be attributed to the manipulated variable. Additionally, the use of standardized task and identical instructions for all participants ensured consistency across conditions, further supporting the reliability of the results. Furthermore, the use of a previously validated questionnaire (USE) to measure satisfaction strengthens the construct validity of the study, as the instrument has been tested and established in prior research (Lund, 2001).

5.3 Limitations

This study has several limitations that should be considered when interpreting the results.

First, the measurements of understanding were based on a set of items specifically developed for this study. While these items were designed to capture participants' understanding of system feedback and task progression, they were not derived from a previously validated scale. This was due to the lack of existing instruments that specifically measure understanding of micro-level interface feedback in short, task-based interactions. Although the items were grounded in established theoretical concepts related to feedback and mental models, the absence of prior validation may affect the reliability and comparability of the results.

Second, the study was conducted using a prototype rather than a fully functional system. As a result, the interaction was limited in realism, and participants' responses may differ from how they would perceive a fully developed web interface in a real-world context. The prototype-based setup may therefore restrict the ecological validity of the findings. In addition, minor design factors beyond the manipulation of microinteractions may have influenced participants' perceptions and should therefore be considered when interpreting the results. Although this allowed for controlled comparison, it may not fully capture how users behave in real-world contexts, where interactions are more complex and prolonged.

Third, the study focused on short, goal-oriented interactions performed during a single session. This means that long-term effects of microinteractions on user experience were

not examined. Consequently, the results reflect immediate perceptions rather than sustained user experience.

In addition, data collection was conducted both in person and remotely via video conferencing. Although efforts were made to standardize the procedure across both settings, differences in testing environments may have influenced participants' experiences. The use of both in-person and remote data collection may have influenced participants' behavior and responses, as the testing environments were not fully standardized. For example, in-person settings allow for greater control over the testing conditions, whereas remote sessions may introduce variations in participants' attention, technical setup, or level of engagement. This variation should be considered when interpreting the results.

Finally, the study is based on a relatively small sample size (n = 20), which limits the statistical generalizability of the findings. While statistically significant differences were identified, a larger sample would be required to confirm the robustness of these results across a broader population. The experimental design allows for controlled comparison between conditions, but the results should be interpreted as indicative rather than definitive and may not be directly generalizable to broader populations or different contexts. Furthermore, the inclusion of participants with design knowledge may have influenced how interface elements were interpreted, which could limit the transferability of the findings.

6 Conclusions and further research

6.1 Conclusions

This study aimed to examine whether and to what extent the presence of microinteractions in a web interface influences users' understanding and satisfaction during short, goal-oriented tasks.

The findings indicate that microinteractions can have a positive impact on both users' understanding and satisfaction during task completion. Participants interacting with the interface including microinteractions generally reported clearer interpretation of system feedback and a more positive overall experience. These results suggest that immediate and context-specific feedback supports users in forming more accurate mental models of system behavior, which in turn contributes to higher satisfaction.

By experimentally isolating the presence of microinteractions, this study contributes empirical evidence to a research area that has previously been largely conceptual. The findings support theoretical perspectives emphasizing the importance of feedback in interactive systems (Norman, 2013) and highlight how even small design elements can influence users' perception of clarity and interaction quality.

From a practical perspective, the results suggest that incorporating microinteractions in interface design can improve users' ability to understand system responses and enhance overall user experience. This is particularly relevant in contexts where users rely on clear feedback to complete tasks efficiently, such as e-commerce environments.

Overall, the study contributes to a deeper understanding of how micro-level interaction design influences user experience, thereby addressing the identified knowledge gap regarding the role of microinteractions in controlled task-based interactions.

6.2 Further research

Future research could build upon this study by addressing several of its limitations. First, larger sample sizes would enable more robust statistical analysis and improve the generalizability of the findings across different user groups.

Second, future studies could examine microinteractions in more ecologically valid environments, using fully functional systems rather than prototypes. This would allow researchers to capture more realistic user behavior and interaction patterns in real-world contexts.

In addition, longer-term studies could investigate how repeated exposure to microinteractions influences users' understanding and satisfaction over time. This would provide insights into whether the observed effects persist beyond initial interactions.

Further research could also explore how different types of microinteractions influence user experience, for example by comparing visual, animated, or interactive feedback elements. This would contribute to a more nuanced understanding of which specific design characteristics are most effective.

Finally, future studies could focus on developing and validating measurement instruments specifically designed to capture understanding of interface feedback. This would strengthen the methodological foundation for studying micro-level interaction effects in human-computer interaction research.

7 References

Bhattacharyya, J. S. Banerjee, & M. Köppen (Eds.), *Human-Centric Smart Computing - Proceedings of ICHCSC 2023* (pp. 13-23). (Smart Innovation, Systems and Technologies; Vol. 376). Springer Science and Business Media Deutschland GmbH. https://doi.org/10.1007/978-981-99-7711-6_2

Boyd, K., & Bond, R. (2021). *Can micro interactions in user interfaces affect their perceived usability?* In Proceedings of the 32nd European Conference on Cognitive Ergonomics (ECCE '21) (Article 40, pp. 1–5). Association for Computing Machinery. https://doi.org/10.1145/3452853.3452865

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77-101. https://doi.org/10.1191/1478088706qp063oa

Davis, F. D. (1989). *Perceived usefulness, perceived ease of use, and user acceptance of information technology*. MIS Quarterly, 13(3), 319–340. https://doi.org/10.2307/249008

Dolić, J., Klepo, H., Pibernik, J., & Mandić, L. (2024). *The influence of microinteractions in registration forms on user experience.* Acta graphica, 32(1), 59–67. Retrieved from https://www.actagraphica.hr/actagraphica/article/view/13/12

Figma. (n.d.). https://www.figma.com/

Hassenzahl, M. (2010). *Experience design: Technology for all the right reasons.* Morgan & Claypool Publishers.

Hestenes, D. (2006). *Notes for a Modeling Theory of Science, Cognition and Instruction.* Arizona State University.

Interaction Design Foundation. (n.d.). *How to use mental models in UX design*. Interaction Design Foundation. https://www.interaction-design.org/literature/article/a-very-useful-work-of-fiction-mental-models-in-design

International Organization for Standardization. (2019). *ISO 9241-210:2019 – Ergonomics of human-system interaction – Part 210: Human-centered design for interactive systems*. ISO. https://www.iso.org/standard/77520.html

Kim, J., Spielmann, N., & McMillan, S. J. (2012). *Experience effects on interactivity: Functions, processes, and perceptions*. *Journal of Business Research, 65*(11), 1543–1550. https://doi.org/10.1016/j.jbusres.2011.02.038

Kolte, A., & Rao, D. (2024). Exploring Microinteractions in Human–Computer Interaction: Design Principles, Types, and User Experience. In S. Bhattacharyya, S.

Lazar, J., Feng, J. H., & Hochheiser, H. (2017). *Research methods in human-computer interaction*.

Lund, A. (2001) Measuring Usability with the USE Questionnaire. Usability Interface, 8(2), 3-6.

Ma, J. Y., & Chen, C. C. (2022). Evaluating user perception and emotion of microinteractions using a contradictory semantic scale. Journal of the Society for Information Display, 30(2), 103-114. https://doi.org/10.1002/jsid.1075

Ma, J. Y., Xie, J. F., & Chen, C. C. (2025). Exploring the structural relationships of microinteractions in perception and behavior by the hedonic motivation system adoption model. International Journal of Human–Computer Interaction, 41(1), 592–607. https://doi.org/10.1080/10447318.2024.2303552

Macefield, R. (2009) *How to Specify the Participant Group Size for Usability Studies: A Practitioner's Guide.* Journal of Usability Studies, 5(1), 34–45. https://uxpajournal.org/how-to-specify-the-participant-group-size-for-usability-studies-a-practitioners-guide/

Maze. (n.d.). *Statistical significance*. Maze. Retrieved February 12, 2026, from https://maze.co/resources/glossary/statistical-significance/

McDaniel, R. (2015). Understanding microinteractions as applied research opportunities for information designers. Communication Design Quarterly Review, 3(2), 55–62. https://doi.org/10.1145/2752853.2752860

Morisso, J. G. D., & Ferreira, M. G. G. (2023). Microinteraction study in user experience: A systematic literature review of research perspectives. In XXVI International Conference of the Iberoamerican Society of Digital Graphics (SIGraDi 2022) (pp. 583–594). Blucher Design Proceedings. https://doi.org/10.5151/sigradi2022-sigradi2022_7

Norman, D. A. (2013). *The design of everyday things* (Revised and expanded ed.). Basic Books.

Pallant, J. (2020). *SPSS survival manual* (7th ed.). McGraw Hill.

Pexels. (n.d.). https://www.pexels.com

Pibernik, J., Dolić, J., Vistid, A., & Mandić, L. (2019). *Time affordance components impact on user's experience.* In Proceedings of the 2019 International Symposium ELMAR (pp. 151–153). IEEE. https://doi.org/10.1109/ELMAR.2019.8918661

Rubin, J., & Chisnell, D. (2008). *Handbook of usability testing.*

Saffer, D. (2013). *Microinteractions: Designing with details*. O'Reilly Media.

Saputra, A., Tarigan, R., & Wijaya, Y. (2024). A Quantitative Study on User Experience Dimensions and Their Impact on User Satisfaction in Indonesian Mobile E-Commerce. *International Journal of Informatics and Information Systems, 7*(3), 135-146. https://doi.org/10.47738/ijiis.v7i3.217

Sundar, S. S., & Kim, J. (2005). *Interactivity and persuasion: Influencing attitudes with information and involvement*. *Journal of Interactive Advertising, 5*(2), 5–18. 10.1080/15252019.2005.10722097

Sutcliffe, A., & Hart, J. (2016). *Analysing the Role of Interactivity in User Experience. International Journal of Human-Computer Interaction.* 33(3). 10.1080/10447318.2016.1239797

Swedish Research Council. (2025). *Good research practice 2024*. Swedish Research Council. https://www.vr.se/english/analysis/reports/our-reports/2025-07-03-good-research-practice-2024.html

Shwany, Z.A.A.A., Sahl, C.H., Abdulrahman, H.A., & Khoshnaw, K.H.K. (2024). *Evaluating The Impact of Micro-Interactions on User Engagement: Study the Effects of Micro-Interactions on User Experiences on University-Level Platforms*. *QALAAI ZANIST SCIENTIFIC JOURNAL*, *9*(2), 1468–1485. https://doi.org/10.25212/lfu.qzj.9.2.54

Yang, S., Linder, J., & Bolchini, D. (2012). *DEEP: Design-oriented evaluation of perceived usability*. *International Journal of Human-Computer Interaction, 28*(5), 308–346. 10.1080/10447318.2011.586320

8 Appendixes

8.1 Appendix 1 – Instruction letter

The following instruction letter was provided to participants prior to the usability test and prototype interaction.

Instructions for user testing

Instruktioner användartester

In this study, we investigate how a website is experienced when performing a simple task. You will be asked to test a prototype of a website.

I denna studie undersöker vi hur en webbsida upplevs när man utför en enkel uppgift. Du kommer att få testa en prototyp av en webbsida.

Your task is to complete a few steps on the website according to the instructions below. There are no right or wrong answers – we are interested in your experience.

Din uppgift är att genomföra några steg på sidan enligt instruktionerna nedan. Det finns inga rätt eller fel – vi är intresserade av din upplevelse.

Task:

Uppgift:

Please complete the following steps on the website:

Genomför följande steg på webbsidan:

1. Mark the product “Relaxed Sweatpants” as a favorite.
2. Add the product “Relaxed Sweatpants” to the shopping cart.
3. Proceed to checkout.
1. *Markera produkten “Relaxed Sweatpants” som favorit.*
2. *Lägg produkten “Relaxed Sweatpants” i varukorgen.*
3. *Gå vidare till kassan.*

After completing the task, you will be asked to respond to a questionnaire about your experience of the website.

Efter att du har genomfört uppgiften kommer du att få svara på en enkät om hur du upplevde sidan.

Your participation is completely voluntary, and you may withdraw at any time without providing a reason. All responses will be treated anonymously and used only for research purposes.

Ditt deltagande är helt frivilligt och du kan när som helst avbryta utan att ange någon anledning. Alla svar behandlas anonymt och används endast i forskningssyfte.

(If recorded:) The interaction may be recorded for analysis, but the material will only be used by the researchers and handled confidentially.

(Vid inspelning:) Interaktionen kan komma att spelas in för analys, men materialet används endast av oss och behandlas konfidentiellt.

By continuing, you consent to participate in the study under these conditions.

Genom att fortsätta samtycker du till att delta i studien under dessa villkor.

Thank you in advance!

Tack på förhand!

8.2 Appendix 2 – Questionnaire

The following questionnaire was administered to participants after completing the task on their corresponding prototype.

This survey concerns your experience of a website you have just tested. The purpose is to understand how different features of the website influence how clear and easy it is to understand, as well as how satisfied you are with the user experience.

Den här undersökningen handlar om din upplevelse av en webbsida du just testat. Vi vill förstå hur olika funktioner på sidan påverkar hur tydlig och lättförståelig sidan känns, samt hur nöjd du är med användarupplevelsen.

The questionnaire takes approximately 5 minutes to complete. Your responses are completely anonymous. Please answer as honestly as possible, both in the multiple-choice questions and in the open-ended questions where you can provide your own comments.

Enkäten tar ungefär 5 minuter att fylla i. Dina svar är helt anonyma. Vänligen svara så ärligt du kan – både på de frågor där du kryssar i alternativ och på de öppna frågorna där du får skriva egna kommentarer.

Thank you in advance!

Tack på förhand!

Which number were you assigned?

Vilket nummer blev du tilldelad?

In the following questions, “the website” refers to the website you just tested.

I följande frågor syftar sidan till den hemsidan du just testade.

Satisfaction (Lund, 2001)

Tillfredsställelse (Lund, 2001)

I am satisfied with the website.

(1 = Strongly disagree, 5 = Strongly agree)

Jag är nöjd med sidan.

(1 = Håller inte alls med, 5 = Håller med helt)

I would recommend this website to a friend.

(1 = Strongly disagree, 5 = Strongly agree)

Jag skulle rekommendera denna sida till en vän.

(1 = Håller inte alls med, 5 = Håller med helt)

The website functions as I expect.

(1 = Strongly disagree, 5 = Strongly agree)

Sidan fungerar som jag förväntar mig.

(1 = Håller inte alls med, 5 = Håller med helt)

The website is pleasant to use.

(1 = Strongly disagree, 5 = Strongly agree)

Sidan är trevlig att använda.

(1 = Håller inte alls med, 5 = Håller med helt)

The interaction with the website felt smooth.

(1 = Strongly disagree, 5 = Strongly agree)

Interaktionen med sidan kändes smidig.

(1 = Håller inte alls med, 5 = Håller med helt)

It was clear what happened when I performed an action on the website.

(1 = Strongly disagree, 5 = Strongly agree)

Det var tydligt vad som hände när jag gjorde en handling på sidan.

(1 = Håller inte alls med, 5 = Håller med helt)

I had a positive experience when completing the tasks on the website.

(1 = Strongly disagree, 5 = Strongly agree)

Jag hade en positiv upplevelse när jag genomförde uppgifterna på sidan.

(1 = Håller inte alls med, 5 = Håller med helt)

I would consider using a similar website again.

(1 = Strongly disagree, 5 = Strongly agree)

Jag skulle kunna tänka mig att använda en liknande sida igen.

(1 = Håller inte alls med, 5 = Håller med helt)

Understanding:

Förståelse

It was easy to understand what happened when I marked the product as a favorite.

(1 = Strongly disagree, 5 = Strongly agree)

Det var lätt att förstå vad som hände när jag favoritmarkerade produkten.

(1 = Håller inte alls med, 5 = Håller med helt)

It was easy to understand what happened when I added a product to the shopping cart.

(1 = Strongly disagree, 5 = Strongly agree)

Det var lätt att förstå vad som hände när jag lade en produkt i varukorgen.

(1 = Håller inte alls med, 5 = Håller med helt)

It was easy to understand how to proceed to checkout.

(1 = Strongly disagree, 5 = Strongly agree)

Det var lätt att förstå hur jag skulle gå vidare till kassan.

(1 = Håller inte alls med, 5 = Håller med helt)

I feel confident that the tasks were completed correctly.

(1 = Strongly disagree, 5 = Strongly agree)

Jag känner mig säker på att uppgifterna har genomförts korrekt.

(1 = Håller inte alls med, 5 = Håller med helt)

Overall, it was easy to understand what happened on the website.

(1 = Strongly disagree, 5 = Strongly agree)

Överlag var det lätt att förstå vad som hände på sidan.

(1 = Håller inte alls med, 5 = Håller med helt)

What worked well on the website and helped you understand what to do? (Open-ended)

Vad fungerade bra på sidan och hjälpte dig att förstå vad du skulle göra?

Was there anything you found unclear that could be improved on the website? (Open-ended)

Finns det något som du tyckte var oklart och som kan förbättras med sidan?

8.3 Appendix 3 – Static prototype

The following images present the full interface of the static prototype used in the experiment.

Figure A1. Homepage of the static interface.

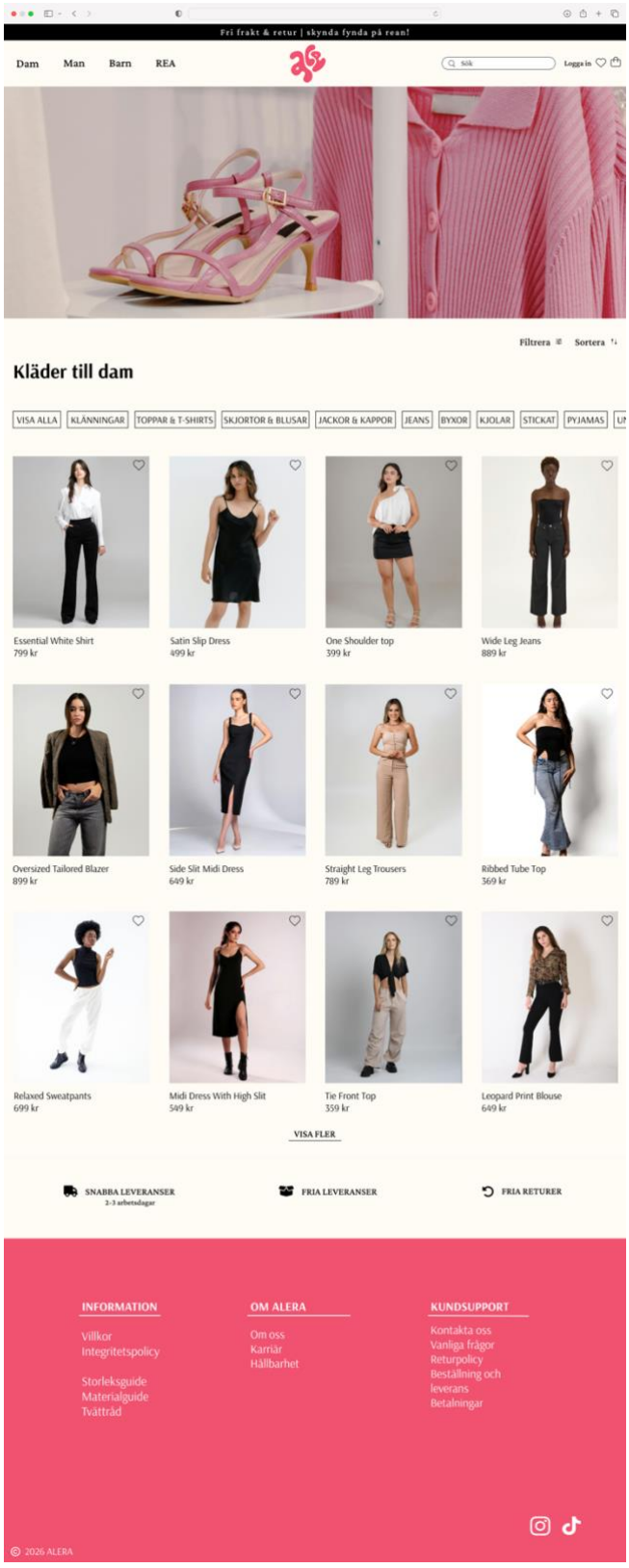


Figure A2. Product page of the static interface.

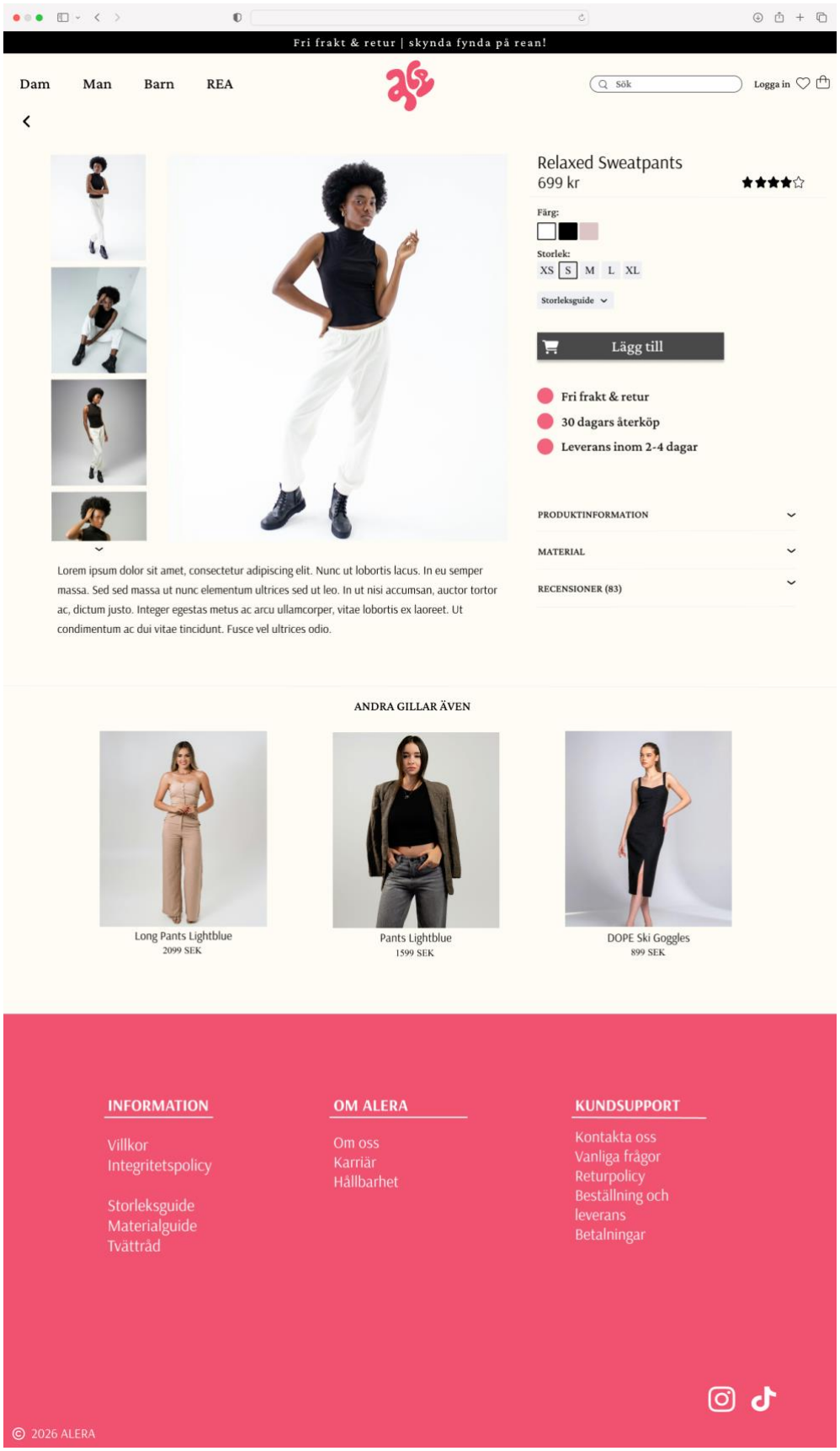
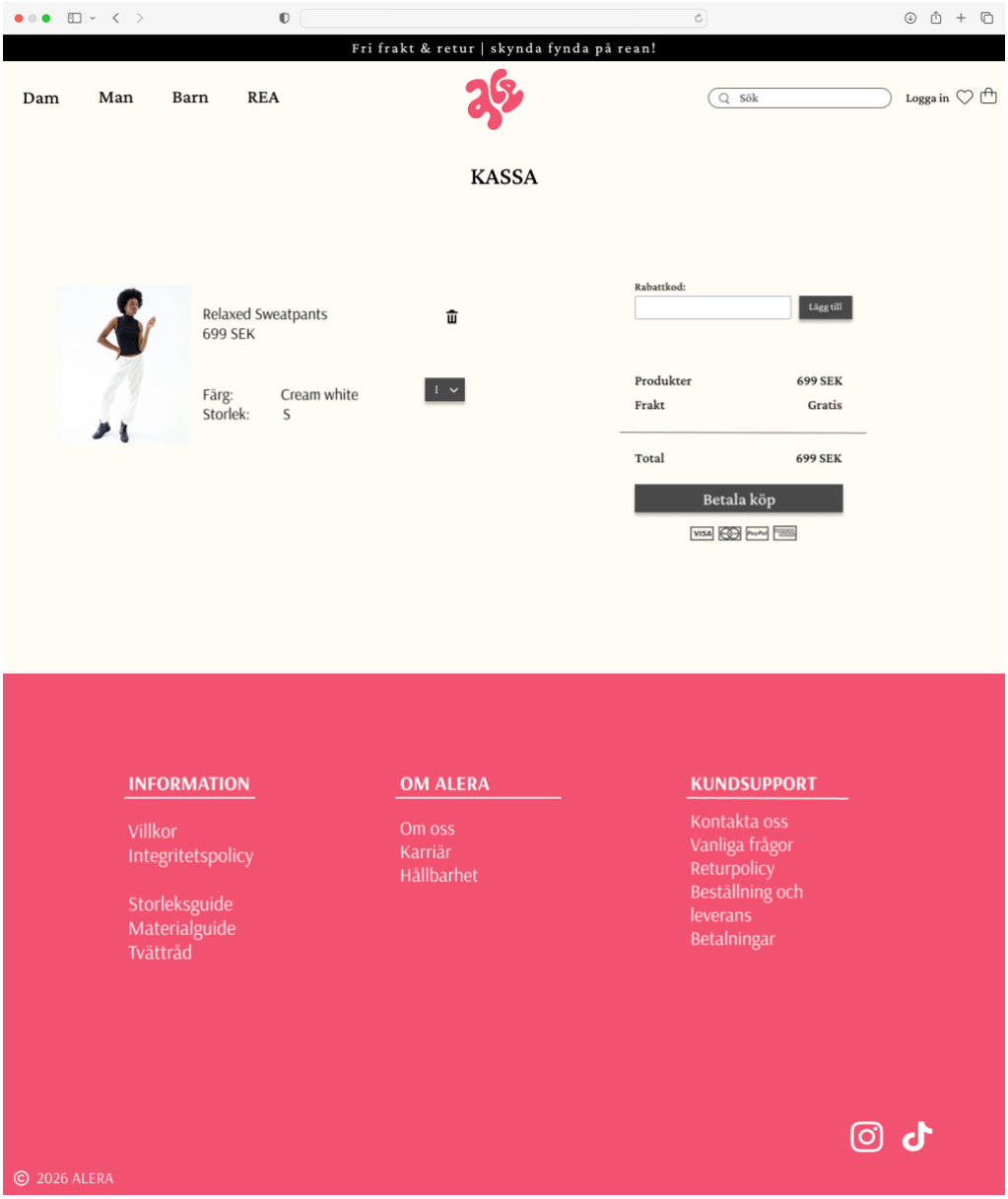


Figure A3. Checkout view of the static interface.



8.4 Appendix 4 – Prototype with microinteractions

The following images present the full interface of the prototype with microinteractions used in the experiment.

Figure B1. Homepage of the interface with microinteractions, after liking the requested product.

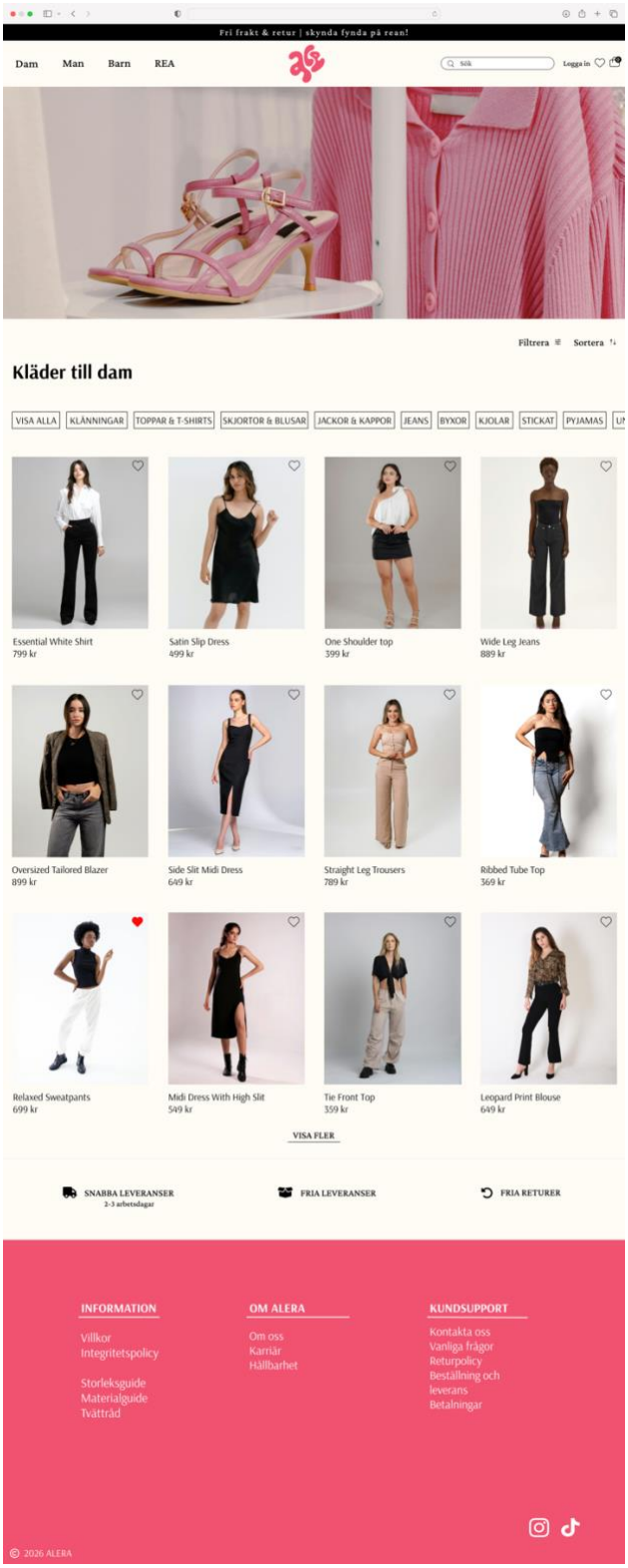


Figure B2. Product page of the interface with microinteractions, after adding the product to the shopping cart.

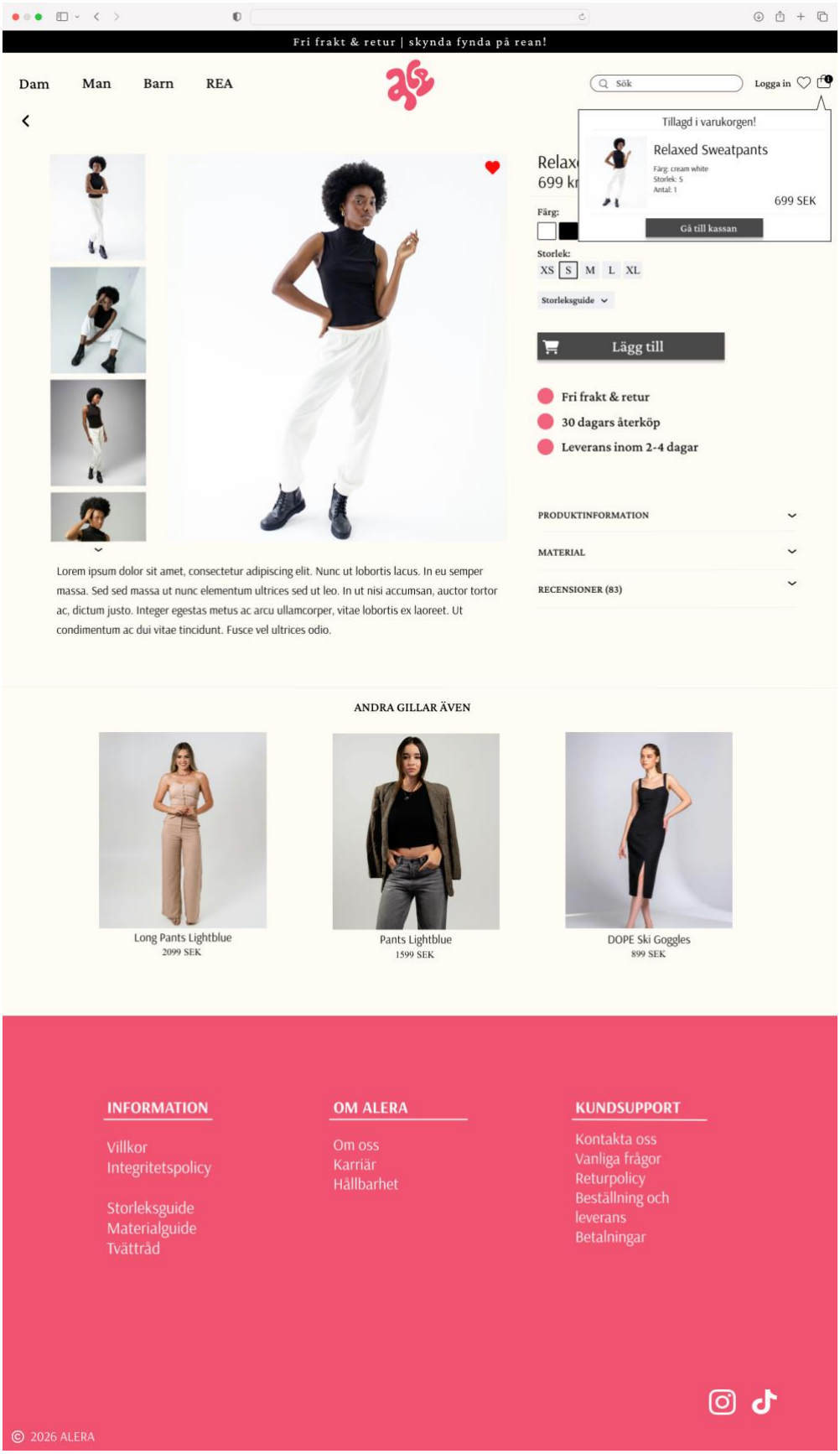


Figure B3. Checkout view of the interface with microinteractions.

